# AMPLENESS PUSHES DOWN ALONG A FINITE SURJECTIVE MORPHISM IN THREE SITUATIONS

THE STACKS PROJECT AUTHORS

*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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*Editorial note.* Statement span 3892-3902 and proof span 3904-3913 come from the halfA pool card operations/pool-cards/stacks-divisors-halfA.json and were re-read in the pinned chapter file before this card was written.

*Editorial note.* The proof is a case-by-case reduction in three short paragraphs, one per hypothesis; the author's enumeration is preserved.

*Editorial note.* The statement's third case writes the characteristic condition as p times the structure sheaf being zero, and the source being the reduction of the target; both are preserved verbatim.

*Editorial note.* 4 cross-references occur inside the counted spans; they are listed in external_references as navigation pointers only.

*Allowed standard prerequisites (editorial).*

- A finite locally free morphism of degree d has a canonical norm of degree d, and the norm of an ample invertible sheaf along a finite morphism admitting a norm is ample. Cited at proof lines 3904-3906, both statements being previous cards of this lane. Gives the first case by composing the construction of the norm with the ampleness statement.
- The Frobenius-like situation gives a norm of degree p for the reduction, and the same ampleness statement applies. Cited at proof lines 3907-3909. Gives the third case, over a base of characteristic p with the source equal to the reduction.
- The norm statement in the normal case, applicable after restricting to an irreducible component dominating the target. Cited at proof lines 3910-3912. Gives the second case.

*Source reference.* Original lemma 0BDZ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14332–14332. Same-chapter reference used in the retained author text at source line(s) 3908. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's

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*Source reference.* Original lemma 0BD0: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14298–14298. Same-chapter reference used in the retained author text at source line(s) 3614, 3781, 3906, 3909. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

*Source reference.* Original lemma 0BD3: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14301–14301. Same-chapter reference used in the retained author text at source line(s) 3912, 3931. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

**Proposition 0.1.** *Let $\pi : X \to Y$ be a finite surjective morphism of schemes. Assume that $X$ has an ample invertible $\mathcal{O}_X$-module. If*

(1) *$\pi$ is finite locally free, or*
(2) *$Y$ is an integral normal scheme, or*
(3) *$Y$ is Noetherian, $p\mathcal{O}_Y = 0$, and $X = Y_{red}$,*

*then $Y$ has an ample invertible $\mathcal{O}_Y$-module.*

*Proof.* Case (1) follows from a combination of Lemmas 0BD2 and 0BD0. Case (3) follows from a combination of Lemmas 0BDZ and 0BD0. In case (2) we first replace $X$ by an irreducible component of $X$ which dominates $Y$ (viewed as a reduced closed subscheme of $X$). Then we can apply Lemma 0BD3. □